EE675: Introduction to Reinforcement Learning

Assignment 1 - Question 3

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1 Working Principles

Question 3 transitions from an environment with a known transition kernel to one where the agent must estimate the model by interacting with a simulator.

1.1 Kernel Estimation (Part a)

The `estimate_kernel` function implements a frequency-based (maximum likelihood) estimation of the state-transition probabilities.

- **Sampling**: For every state-action pair (s, a), the simulator is called K times.
- **Empirical Frequency**: The transition probability $\hat{P}(s'|s, a)$ is calculated as the ratio of times the simulator resulted in state s' divided by the total samples K.
- **Sparse Implementation**: Given the N^4 state space, we use a COO-to-CSR conversion. This allows us to accumulate "probabilistic weights" $(1/K)$ for each observed transition efficiently without manually managing counts.

1.2 Estimated Q-Function and Error Analysis (Part b)

Using the estimated kernel $\hat{\mathcal{P}}$, we apply the previously developed Value Iteration algorithm to find the approximately optimal Q-function, $\hat{Q}$.

- **Metric**: The accuracy of the estimation is measured using the L_1-difference against the exact optimal Q-function (Q^*) obtained in Question 2: $L_1 = \sum_{s,a} |Q^*(s, a) - \hat{Q}(s, a)|$.
- **Convergence**: As K increases, the Law of Large Numbers ensures $\hat{\mathcal{P}}$ approaches the true $\mathcal{P}$, resulting in a lower L_1 error.

1.3 Statistical Robustness (Part c)

To account for the inherent randomness in the predator-prey simulator (specifically the uniform movement of the prey), the experiment is repeated across five independent random seeds.

- **Mean and Variance**: We compute the mean L_1 error to observe the general trend and the standard deviation to measure the estimator's sensitivity to simulation noise.

2 Naming Conventions

- `estimate_kernel.py`: File containing the model estimation logic.
- `K`: The number of samples per state-action pair (input to the estimator).
- `hat_kernel`: The estimated transition kernel matrix ($\hat{\mathcal{P}}$).
- `Q_optimal_exact`: The "Gold Standard" Q-function solved using the exact kernel.

3 Results

3.1 Single Estimation Error and Statistical Variance

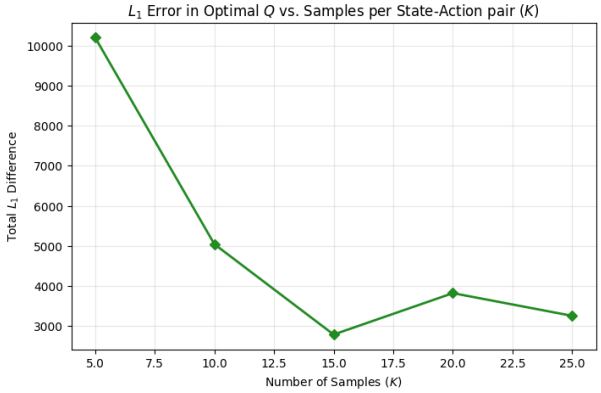


Figure 1: L_1 error vs K (Single Seed).

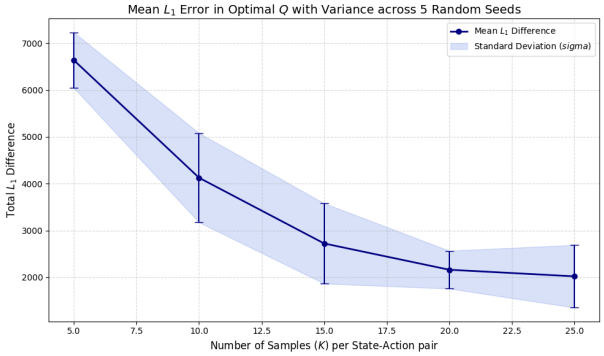


Figure 2: Mean and Std Dev over 5 Seeds.

The plots demonstrate that increasing the sampling budget K significantly reduces the error in the learned Q-function. The reduction in standard deviation (shaded region) further confirms that higher values of K lead to a more stable and robust model estimation.